Faculty of Information Communication Technology

Module Name: Process testing
Module code: PCT316D
Lecturers: Dr Segooa, Ms Ntsoane, Mr Mashego, Ms Hutang
Submission Due Date: **5 May 2026 11:59 PM**
Mode of submission: Brightspace, and the group leader sends a copy via email.
Presentation Date: 08 May 2026 (10:00-12:00)
Total Mark = 100

Read the following scenario and answer the project questions.

Testing is one of the most important parts of developing a new software product or solution. Different techniques may be applied to improve the quality of the solution. Seeing that we are in the fourth Industrial Revolution there are emerging tools and technologies that are suggested to influence the efficiency of most businesses. Assume that you are appointed to work on a project as a tester and your team is requested to assist in solving the SME's challenges in the community.

The project will employ a **System Development Life Cycle (SDLC)** methodology (planning, analysis, design, development, testing… as a structure and the application of test principles and test processes.

Create a Google doc to document your project and send a link to the respective lecturer. This is a live document you will work on throughout the project.

1. **Discuss the new trends in software testing in this Digital Era. (10)**
2. **Visit one small and medium enterprise (SME) within a community that you would like to serve through your project. During the visit, you need to establish the business challenges (Observations and interviews can be done) [Requirements gathering]. (10)**
3. **Based on the visit and observations, establish the challenges that SME encounters and define the problem [Requirements analysis]. (10)**
4. **Based on the problem defined, what do you think may be solutions to the problem? You may come up with various ideas of solutions and decide on the best suitable solution [Solution analysis]. (10)**
    - Use three AI tools of your choice to **identify** three possible **solutions** to the problem. (3)
    - **Propose** the best possible option that may be used to address the problem (3)

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- **Summarise** benefits (Value) (2) and **Context** (2) in which the proposed solutions can work (tabular form).

5. **Design the technology based on the SMEs/users' needs [Design]. (10)**
6. **Collaborate with AI to develop/outsource the software solution, this can be something as simple as a customer registration Form. The Enterprise name and AI tool used should appear on the form. [Development]. (20)**
7. **Construct/Formulate/create test scenarios for the solution you came up with. [Testing]. (5)**
8. **Convert the test scenario into test cases [Testing]. (10)**
9. **Test the solution using the test case created in Q8 and share the test report [Testing]. (10)**

Guidelines for submission

The project must be the author's original work. No plagiarism will be tolerated! (NO COPY AND PASTING)

- This is group work.
- Nominate a group leader.
- Work collaboratively as a team.
- First, work individually, document your responses, then share with the team during